Rion Oh
(she/her)

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Department of Physics, KAIST, Daejeon 34141, Republic of Korea

RESEARCH INTERESTS

Early galaxy formation and evolution, baryon cycle, star formation histories, quiescent galaxies, post-starburst galaxy, dusty star-forming galaxies, galaxy lensing

EDUCATION

KAIST (Korea Advanced Institute of Science and Technology) — Daejeon, South Korea
B.S. in Physics, Minor in Electrical Engineering — Expected in Aug 2027
Thesis: The Evolutionary Connection Between Dusty Star-Forming and Quiescent Galaxies at $z = 2$–6

University of Washington — Seattle, WA
Exchange Student; Research and coursework in the Department of Astronomy — Jan 2026 –Aug 2026

PUBLICATIONS

1. **Oh, R.** et al. *Spatially Resolved Neutral Gas Outflows in Lensed Galaxies at z=3–5*. 2026, in prep
2. **Oh, R.** et al. *Statistical Detection of Neutral Gas Outflows in $z = 2$–5 Quenching Galaxies with JWST*. 2026, in prep
3. Khullar, G., ..., **Oh, R.**, et al. *LEGGOS I: The JWST LEGGOS Survey – LEnsing and Galaxy Growth: Observing Substructures – Unpacks the Nature of Clumpy Star Formation and Quenching in Gravitationally Lensed Galaxies beyond Cosmic Noon*. 2026, arXiv:2606.20845

RESEARCH EXPERIENCE

Neutral Gas Outflows in High-Redshift Quenching Galaxies — Dec 2025–present
Red Galaxies Group, University of Washington, *Advisor: Prof. Arianna S. Long*

- Performed Prospector SED fitting and spectral stacking of 274 galaxies at z=2–5 to Na I D absorption
- Detected neutral gas outflows in all quiescent bins, with outflow properties 0.5–2 dex above non-quiescent
- Showed outflow energetics exceed stellar feedback, implicating AGN-driven quenching

Spatially Resolved Neutral Gas Outflows in Lensed Galaxies — Apr 2026–present
JWST LEGGOS Team, *Advisor: Dr. Gourav Khullar*

- Extracting spatially resolved, clump-level spectra from JWST NIRSpec observations of 4 lensed galaxies
- Measuring Na I D, Mg II, and Fe II absorption to constrain neutral gas flows across individual clumps
- Connecting clump-level gas kinematics to physical properties and evolutionary phase

The DSFG-to-Quiescent Transition at $z = 2$–6 — Dec 2025–present
Observational Cosmology & Astrophysics Laboratory, KAIST, *Advisor: Prof. Junhan Kim*

- Classified 249 galaxies at $2 \leq z \leq 6$ and identified 24 DSFGs via Prospector dust attenuation ($A_V \geq 1$)
- Showed DSFGs span diverse star-formation phases despite similar A_V and mass
- Identified 2 PSB candidates at $z > 3$, suggesting rapid quenching can occur within the DSFG phase

Triaxial Modeling of Intracluster Gas with X-ray and SZ Observations — Jun–Dec 2025
Observational Cosmology & Astrophysics Laboratory, KAIST, *Advisor: Prof. Junhan Kim*

- Modeled intracluster gas using joint X-ray and SZ observations within the CLUMP-3D framework
- Implemented triaxial density and pressure models projected along the line of sight
- Performed Bayesian MCMC inference on a 13-parameter cluster model

AWARDS & HONORS

Dean's List, *University of Washington* — Winter, Spring 2026

Korea–U.S. Advanced Fields Youth Scholar, *KIAT & IIE*, national selection (200 awardees) — 2026

Astra Recon Award (satellite), Hanwha Aerospace &Systems Space Challenger, top 5 teams — 2025

Outstanding Student for Academic Inquiry, *KAIST Physics*, 2 students (astrophysics) — Fall 2025

2nd Place, Scientific Writing Award (English), *KAIST* — 2024

Academic Excellence Scholarship, *KAIST*, merit-based — 2024–2027

Institutional Merit Scholarship, *KAIST*, full-tuition — 2023–2027

Science & Technology Talent Scholarship, *KAIST*, merit-based — 2023

PRESENTATION

Extragalactic Astronomy Symposium,*University of Washington* — Apr 2026
Oral: "Statistical Detection of Neutral Gas Outflows in $z = 2$–5 Quenching Galaxies with JWST"

Undergraduate Research Symposium,*University of Washington* — Apr 2026
Poster: "Statistical Detection of Neutral Gas Outflows in $z = 2$–5 Quenching Galaxies with JWST"

RESEARCH TRAINING & PROJECTS

Extragalactic Astronomy & Cosmology Summer School — Jul 2025
Korea Institute for Advanced Study

- SIDM cosmological simulations with *MUSIC* & *GIZMO*: core fraction analysis as a function of halo mass
- Galaxy merger simulations with *DICE* & *GIZMO*: dark matter, gas, and stellar dynamics
- Radio telescope receiver circuit design; antenna parameter constraints for 21 cm intensity mapping

Radio Summer School & Radio Telescope Users' Meeting — Aug 2025
Korea Astronomy and Space Science Institute

- Constructed and operated a small radio telescope; performed data acquisition and analysis
- Visited KVN data center; attended Users Meeting covering major facilities (ALMA, EAVN, VLBI)

Stratospheric Balloon Payload Project, *KAIST* — Feb-Nov 2025

- Built a radiosonde payload with custom circuitry and sensor integration
- Modeled trajectories to predict landing coordinates across two launches

OBSERVATION

Apache Point Observatory 3.5m Telescope, KOSMOS Observing Run (5 hours) — May 14, 2026

- Spectroscopic follow-up of lensing candidates using g-band emission lines for redshift constraints

NYSC 1m Telescope, Deokheung Optical Astronomy Observatory (6 hours) — Apr 01, 2025

- Time-series photometry of Kepler-17b transits; extracted light curves and derived orbital parameters

TEACHING EXPERIENCES

Instructor, Elementary STEM Outreach Program — Feb–Jun 2025

Instructor, Secondary-Level Physics — Mar–Dec 2024

Tutor, Undergraduate STEM Subjects (General Physics, Calculus, General Chemistry) — Feb 2023–Dec 2025

SERVICES & OUTREACH

Founder & Author, *The Young Universe* Newsletter — Apr 2026–present

- Korean & English high-redshift galaxy paper review newsletter — `maily.so/theyounguniverse`

Mentor, Program for Deaf Youth — Feb 2024–present

Student Council Member, Department of Physics, KAIST — Feb 2024–Jan 2026

Department Representative, Department of Physics, KAIST — Feb 2024–Jan 2025

REFERENCES

Prof. Arianna S. Long, Department of Astronomy, University of Washington (aslong@uw.edu)
Dr. Gourav Khullar, Department of Astronomy, University of Washington (gkhullar@uw.edu)
Prof. Junhan Kim, Department of Physics, KAIST (junhan@kaist.ac.kr)